

Aspect-Based Sentiment Analysis of Code-Mixed Data Using a Transliteration-Aware BiLSTM-Attention Model

Saurabh Tripathi
M251020CS

Guided by: Pournami P.N.

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Abstract

Sentiment analysis is a key task in Natural Language Processing (NLP) that focuses on identifying and classifying opinions expressed in text into sentiment categories such as positive, negative, or neutral. With the exponential growth of social media, online review, large volumes of user-generated content are produced daily, making automated sentiment understanding essential for applications like opinion mining and recommendation systems [1, 2, 3]. However, social media, online reviews text is often noisy and informal, containing slang, emojis, and multilingual expressions, which makes accurate sentiment classification challenging.

This research focuses on Aspect-Based Sentiment Analysis (ABSA) of Hinglish code-mixed text, where Hindi and English are mixed in Roman script. The proposed approach incorporates a transliteration-aware preprocessing module to normalize inconsistent Romanized Hindi words, followed by embedding generation and a Bidirectional Long Short-Term Memory (BiLSTM) network to capture contextual dependencies [4]. An attention mechanism is used to focus on sentiment-relevant words at the aspect level, improving classification performance for positive, negative, and neutral sentiments.[5].

1 Introduction

Natural Language Processing (NLP) is a subfield of artificial intelligence that enables machines to understand, interpret, and generate human language in a meaningful way [3]. It combines computational linguistics, machine learning, and statistical methods to process and analyze large-scale textual data. NLP plays a crucial role in applications such as machine translation, question answering systems, chatbots,

and information retrieval, where understanding human language is essential [1, 2]. With the rapid expansion of digital communication platforms, NLP has become increasingly important for extracting useful insights from unstructured text data such as social media, online reviews, posts, product reviews, and online discussions.

Despite significant advancements, NLP systems face multiple challenges when dealing with real-world textual data. Social media, online reviews content is highly unstructured and often includes slang, abbreviations, emojis, spelling variations, and grammatical inconsistencies. These factors make it difficult for traditional models to accurately interpret meaning. Additionally, sarcasm, ambiguity, and context-dependent expressions further complicate language understanding. Multilingual and code-mixed text adds another layer of complexity, where multiple languages appear within the same sentence, often without consistent grammar or structure [6, 7].

To address these challenges, this work focuses on Aspect-Based Sentiment Analysis (ABSA), which aims to determine sentiment polarity with respect to specific aspects mentioned in a sentence. Unlike general sentiment analysis, ABSA provides fine-grained opinion mining by associating sentiment with particular entities or features. This makes it more suitable for analyzing real-world user feedback, where a single sentence may contain multiple opinions about different aspects [8, 9].

In this research, we specifically target Hinglish code-mixed text, where Hindi and English are mixed in Roman script. Such data is widely observed on online reviews and presents significant challenges due to inconsistent transliterations and non-standard spellings. For example, the same Hindi word may appear in multiple forms, such as “acha”, “achha”, or “accha”, leading to vocabulary sparsity and ambiguity.

To handle this issue, we propose a transliteration-aware preprocessing module that normalizes noisy Romanized Hindi words into a consistent form before further processing. This step reduces lexical variation